Impulse On-Site

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PURDUE UNIVERSITY

Contents

1 **Background & Experience**
Avionics, Test Engineering, Leadership

2 **Project Overview**
Problem Statement & Motivation

3 **Development**
Revisions, Challenges and Timeline

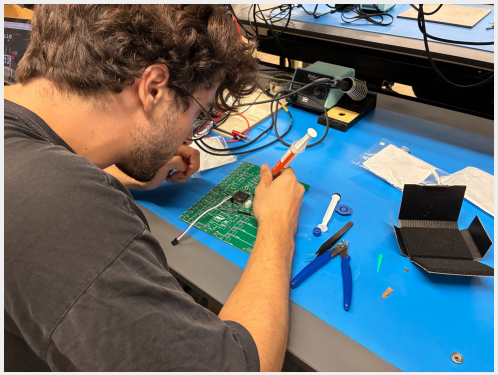
Background & Experience

Purdue University:

- BS in Electrical Engineering - December 2026
- MS in Electrical Engineering - December 2027

Purdue Space Program AC Avionics Lead

- Led team from January 2026 to current day
- Developed Engine Controller PCB
- Oversaw two test campaigns, one more this semester



ASML Production & Test Engineering Intern

- Developed automated coil resistance checker
- Participated in cleanroom escalations

PSP Active Controls

- **Mission**
  - Build a liquid rocket for autonomous 50m takeoff, hover, and landing of lander vehicle
- **Our Path**
  - ASTRA - VTOL drone used for GNC testing
  - TADPOLE - 550 lbf rocket engine + DAQ system
  - **DART** - Custom Igniter
  - **SKIPPER** - Regeneratively Cooled Rocket Engine
  - TOAD - the whole rocket
- **My Role**
  - End to end design of controller for **Skipper**
  - Responsible engineer for **DART** Spark Box
  - Leading requirements development for TOAD

ASTRA



TADPOLE



DART



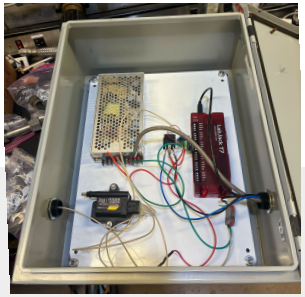
SKIPPER



TOAD

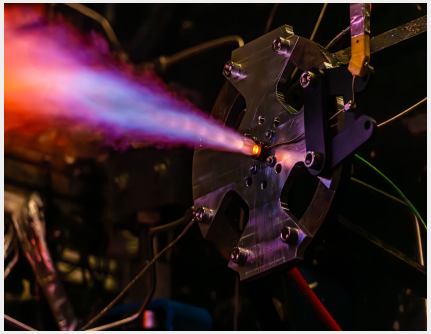


PSP Projects





- **Engine Controller** - Valve controller and DAQ
- **DART** - Electrical Box supplying spark coil with 40kV
- **PT (Pressure Transducer) Redesign** - Simplified existing TADPOLE wire PT ADC Boards
- **TADPOLE Wire Harnessing** - Soldered, Crimped, a LOT
- **Zucrow DAQ Wiring** - Remade old cabling for AI, DO wires at Zucrow Test Stand A
- **Onboarding** - Designed onboarding for new members, teaching Altium, LTSpice and KiCAD



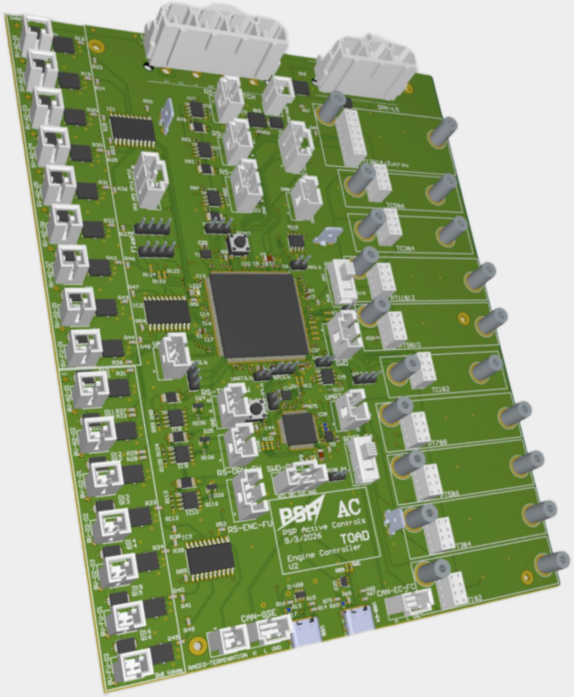


Project Overview

Engine Controller

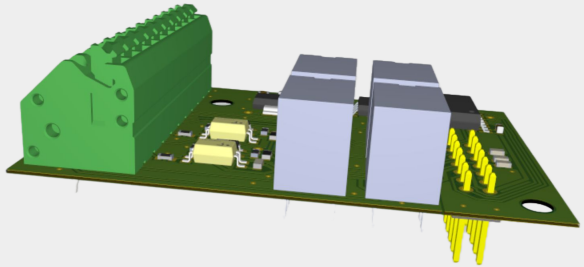
- What does a 550 lb liquid rocket need for a successful flight that ASTRA (the drone) does not?

- My contributions to this project:
  - Requirements
  - Design Reviews
  - PCB Design
  - PCB Layout
  - Buildup
  - Test



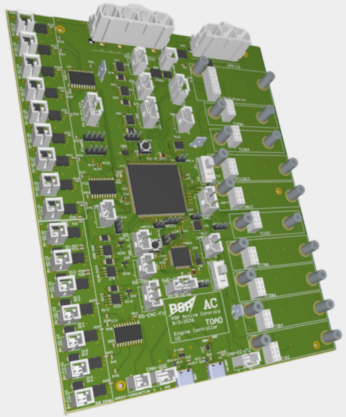
Functionality

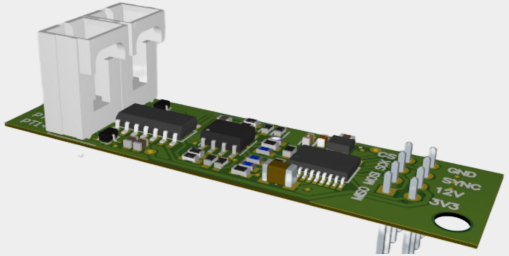
1. Controls Valves (Ball, Solenoid, and Throttle)
2. Read pressure & temperature data
3. Control spark coil
4. Interface with Zucrow



Modular design

- Engine Controller
- PT ADC Boards
- TC ADC Boards
- Zucrow DAQ Interface





Development

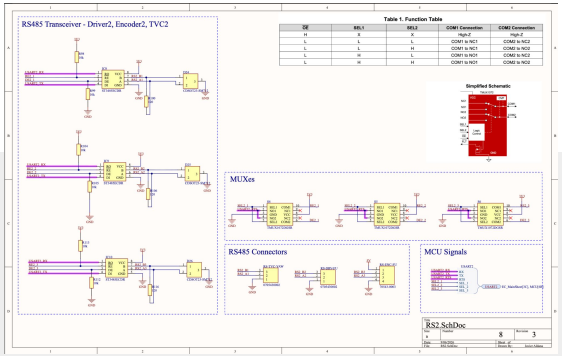
Requirements

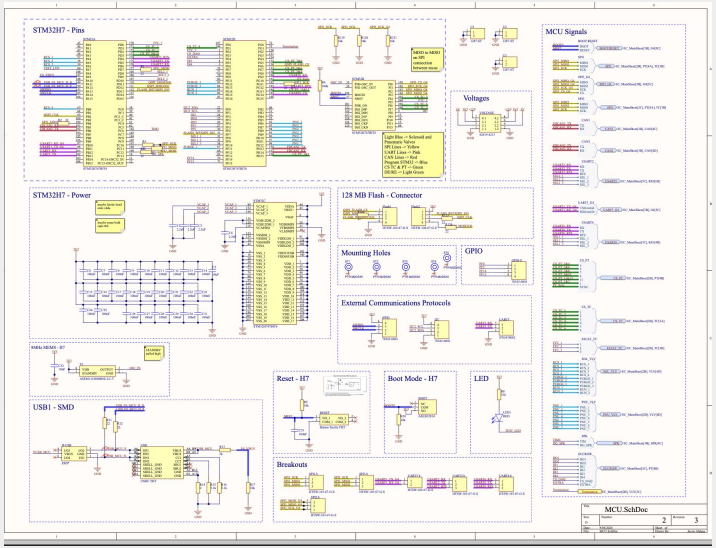
Requirements	Numbers	Reasoning
Pressure & Temperature	15 PT, 5TC	Dictated by PID
Spark	1, 12V @ 8A	Requires 96W and 40kV at the spark plug.
Valves	7 Ball, 9 Solenoid, 2 Throttle Valves	Dictated by PID
Communication Protocols	2 CAN, 6 RS485, 2SPI	Stepper motors, encoders require RS485 Linear Actuators require CAN FD, RS485 ADCs require SPI
Flashing MCU from distance	1	Code MCU while staying safe
Battery	36V, 21Wh, 600mAh	All power for TOAD
Size, Weight	30 lbs, 12”x20”	Keep TWR above 1.4

Design

- Designed in **Altium**
- **2 Versions**
  - ECv1: Prototype
  - ECv2: Mission Viable Product

- **Hierarchical Schematic**
  - MCU - Main Microcontroller
  - G4 - STM32G4 Programmer
  - SPK - Spark
  - PT - Pressure Transducer ADC (Analog to Digital Converter)
  - TC - Thermocouple ADC
  - RS485_1 & 2 - RS485 Transceivers
  - CAN - CAN FD Transceivers



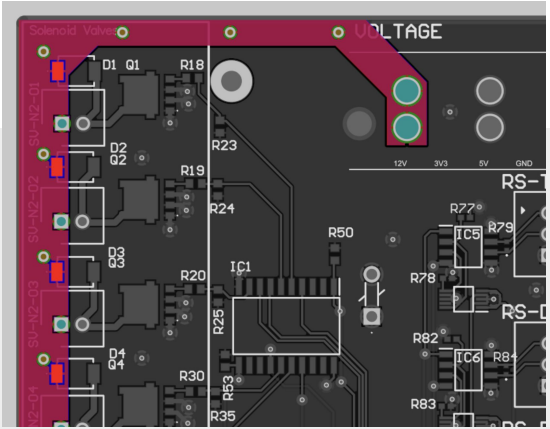


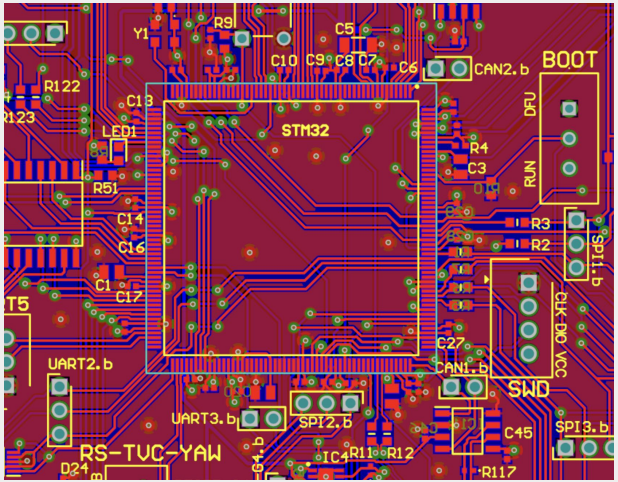
Components

Components	Amount	Reasoning
STM32h7	1	Easy for sw, high pin count, fast
STM32G4	1	Smaller pin count, necessary peripherals
SPARK NMOS: SIR800DP-T1-GE3	2	Rated for 12V at >> 10A, Vth of 600mV
VALVE NMOS: NVMFS5C442NLAFT1G	16	Rated for 12V at >>0.71A, Vth of 2V
CAN FD: MCP2542FD-E_SN	2	8MHz allows for tighter control loops
RS485 Transceivers: ST3485ECDR	6	Runs off of 3.3V, 12Mbps, 1uA shutdown mode

Layout

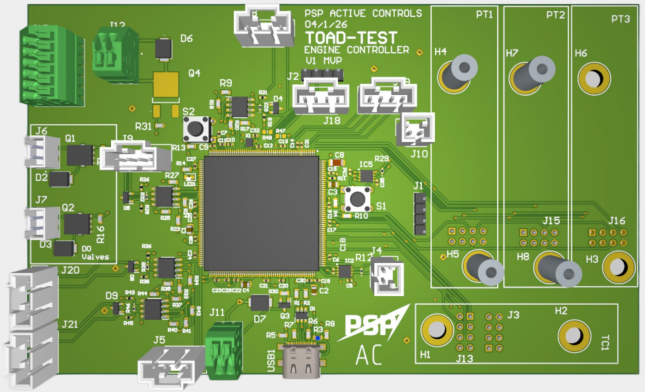
- 4 Layer: Signal, GND, 3V3, Signal
  - ECv1: 3.5”x 5.9”
  - ECv2: 8”x6”
- Differential routing on USB, CAN
- Breakouts for communication signals
- Rated for 8A @ 12V on Solenoid and Ball Valves, rated for 8A@ 12V on Spark
- 302 Components total on ECv2
  - 91 Resistors
  - 46 Capacitors





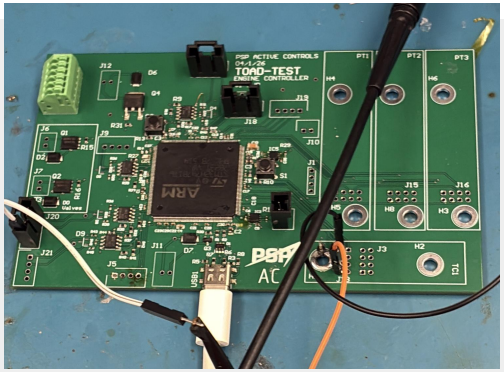
Engine Controller Version 1 (ECv1)

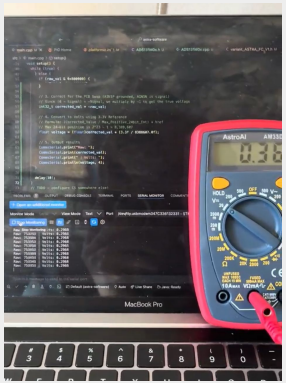
- Proof of concept, **NOT** designed to be mission viable
    - STM32h7 main MCU dual core
    - 3 PT, 1 TC port
    - Spark Coil
    - 1 RS485 transceivers
    - 1 CAN transceiver
    - 1 Solenoid Valve switch
    - 1 Ball Valve switch

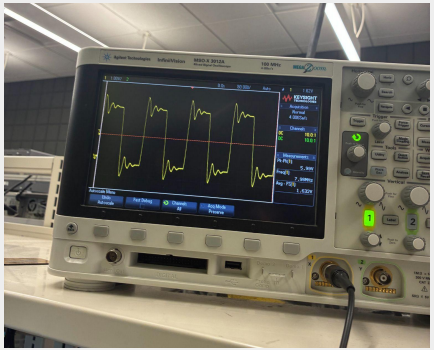


ECv1 Bring-up and Validation

- Soldered SMD components w hot plate
- Flashed over USB
- Tested SPI line with PT and TC boards on
- Debugged noisy clock line
- Tested lowside switch
- Poor labeling





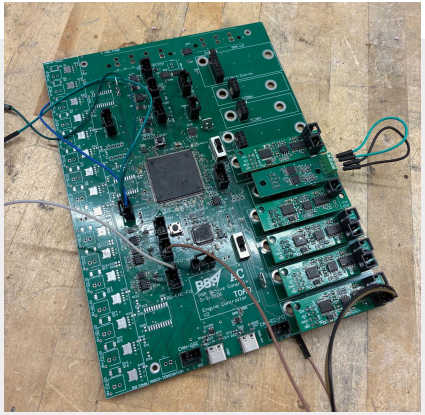


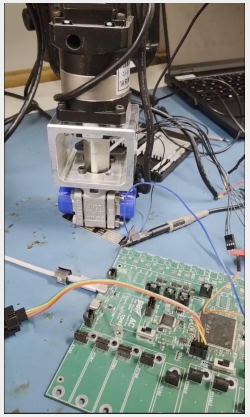
Engine Controller Changes

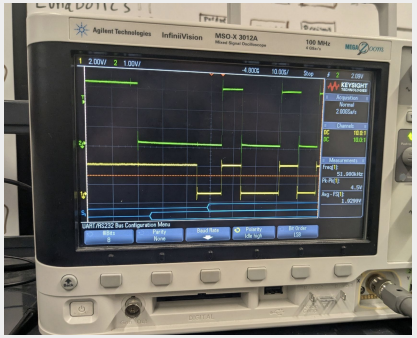
Version 1	Version 2
3 PT boards, **1** TC Board	**8** PT Boards, **3** TC Boards
1 Ball Valve, **1** Solenoid Valve	**9** Ball Valves, **7** Solenoid Valves
1 CAN Transceivers	**2** CAN FD Transceivers
1 RS485 Transceiver	**6** RS485 Transceivers, **6** MUXes
0 programmer boards	**1** STM32G4 to flash main MCU
No communication with Zucrow	**1** sub-board to talk w/ Zucrow DAQ
2MB onboard STM32h7 Flash	Extra 512MB Flash

ECv2 Bringup

- Bringup
  - Soldered main MCU, decoupling caps, flashed over SWD
  - Soldered SMD components, THT (except valves)
  - Flashed code to G4 and H7 (SWD and USB)
- Test so far
  - RS485 Muxing, throttle valve control
  - PT, TC board connection over SPI
  - Integrating w Zucrow DAQ

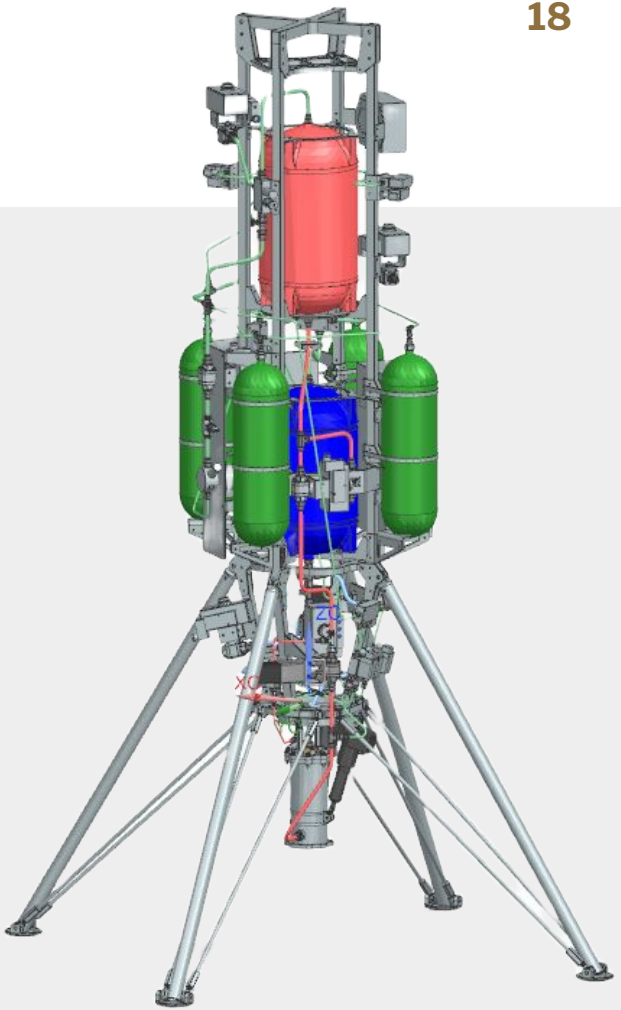




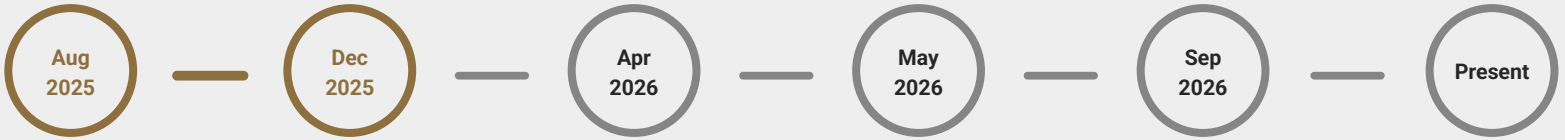


Next Steps

- Spark test
- Testing **signal integrity** on ADC inputs
- Flash code to H7 over SPI from G4
- Wire Harnessing
- Testing full **throttle control**
- **Cold Flows, Hotfire** for SKIPPER engine mid November
- ECv3 will start to be worked post SKIPPER Campaign



Timeline



Requirements

Initial ideation, requirements Research

iPDR

Internal Preliminary Design Review

ECv1 Ordered

Engine Controller Version 1 (ECv1) schematic capture, layout

ECv1 Bringup

Engine Controller Version 1 assembled, tested

ECv2 Ordered

Engine Controller Version 2 (ECv2) schematic capture, layout

Testing

Engine Controller Version 2 assembled, currently being tested

What I bring to the team

- End to End Project Experience
  - SKIPPER **Engine Controller**
  - DART Igniter **Spark Box**
- Technical Knowledge
  - Experience with **scopes, waveform generators**, etc.
  - Soldering, crimping, wire harnessing
  - **Altium, LTSpice,** KiCAD
  - Classes include Power Electronics, Microcontrollers, Robotics, Digital Signal Processing
- Leadership
  - Grown Avionics team **from 2 people to 11**
  - Developed several young engineering students

Why Impulse? Why EP?

- **Culture:** Young, high stakes, fast.
- **Project Ownership:** Small team allows me to take ownership.
- **Power Electronics:** All time favorite class at Purdue. (ECE 31033)
- **Friends:** Many of my close friends are working at this company!





Thank You!

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